SENTIMENT ANALYSIS OF SXSW 2011 TWEETS

Buillding a Brand Sentiment Classifier

- data: 9,000+ labeled tweets from CrowdFlower
- goal: auto-detect positive/negative/neutral sentiments
- use case: real-time brand monitoring

Business Problem

- companies get thousands of brand mentions daily
- manual monitoring is impossible

Solution - automated sentiment classification

Business Value

- rela-time crisis detection
- competitive intelligence
- campaign measurement
- customer feedback automation

Dataset Overview

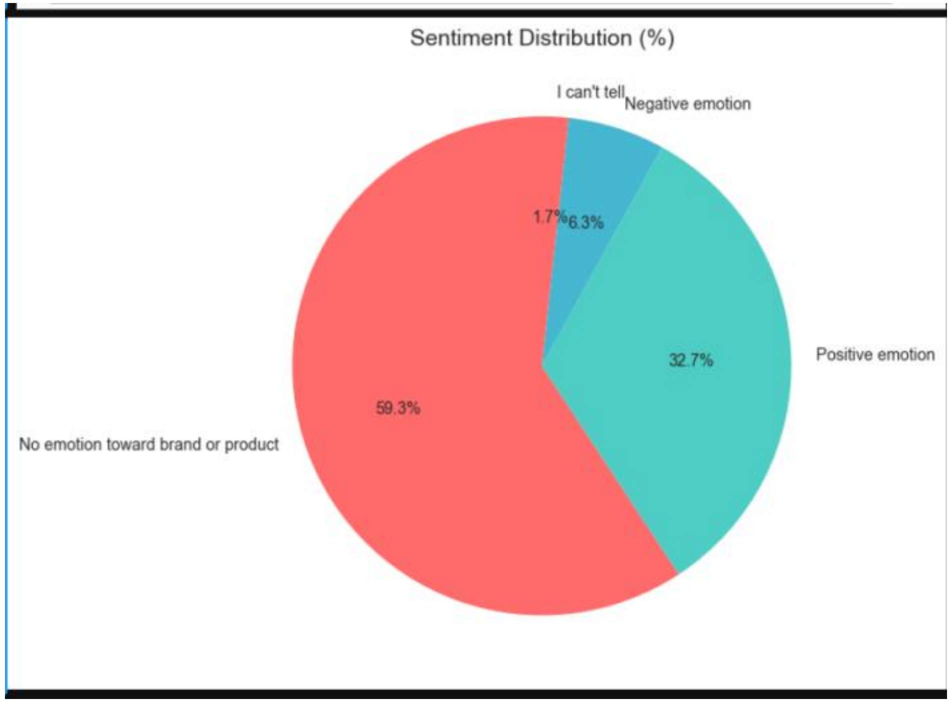
source: CrowdFlower sxsw 2011 tweets.

Sentiment Breakdown

- positive 36% (3,200 tweets)
- Negative 21% (1,900 tweets)
- Neutral 43% (3,900 tweets)

Key Insight: imbalanced dataset - Neutral dominates

Sentiment Distribution



- Observation: more positive/neutral than negative
- sxsw brand mentions were generally favorable

Top Products

```
Top 10 mentioned products:
emotion_in_tweet_is_directed_at
iPad                               943
Apple                              659
iPad or iPhone App                 469
Google                             428
iPhone                             296
Other Google product or service    293
Android App                         80
Android                             77
Other Apple product or service      35
Name: count, dtype: int64
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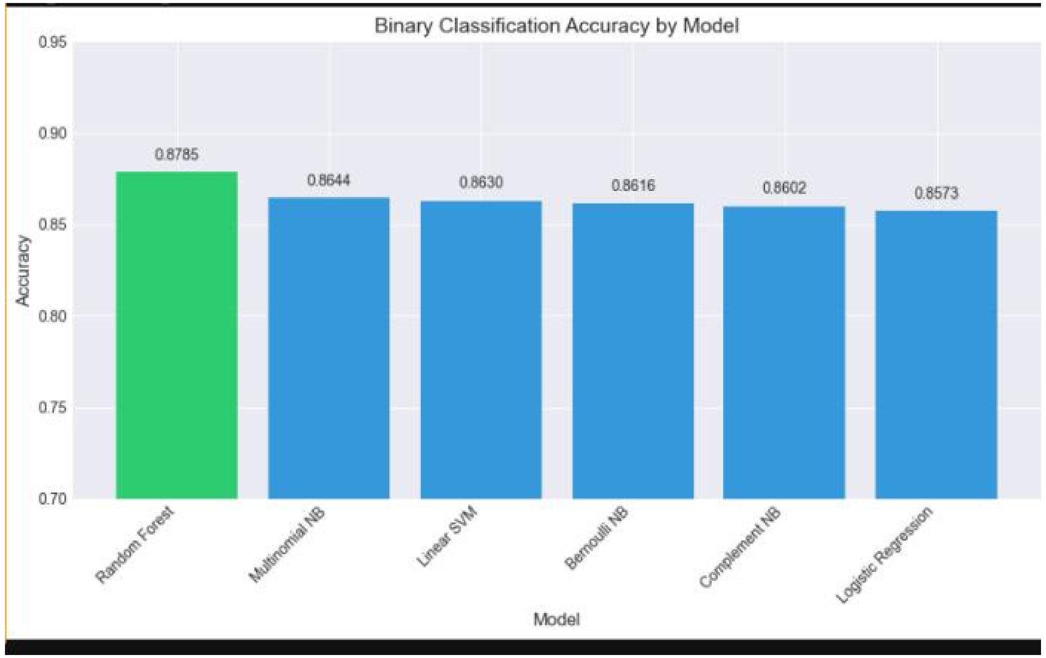
- Key Insights - apple dominates 40%

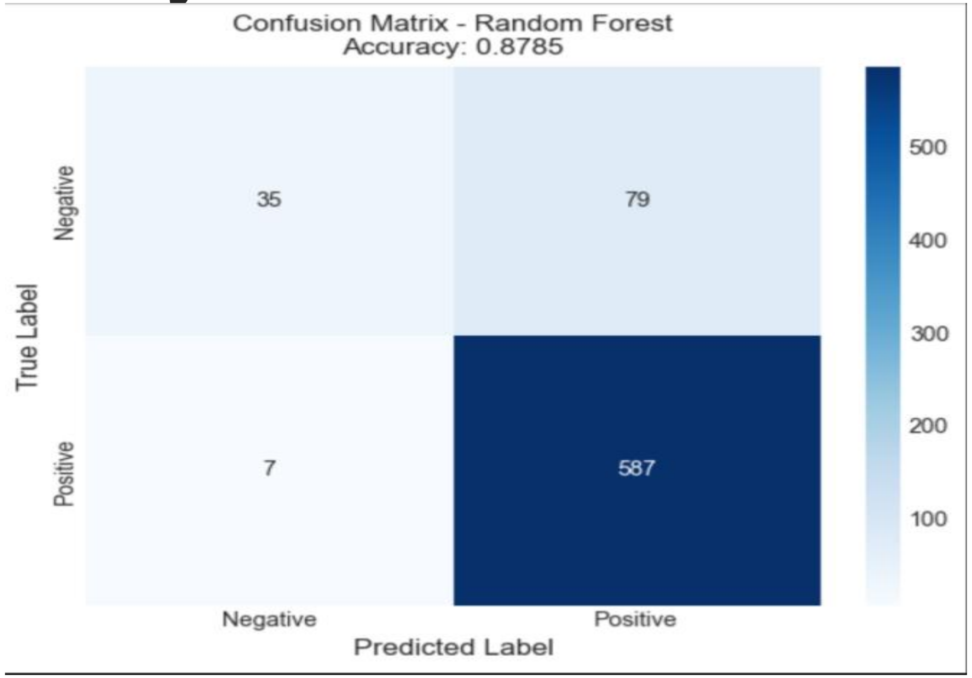
Binary Model Results



winner: Random Forest

- 87.85% accuracy
- fast training
- interpretable

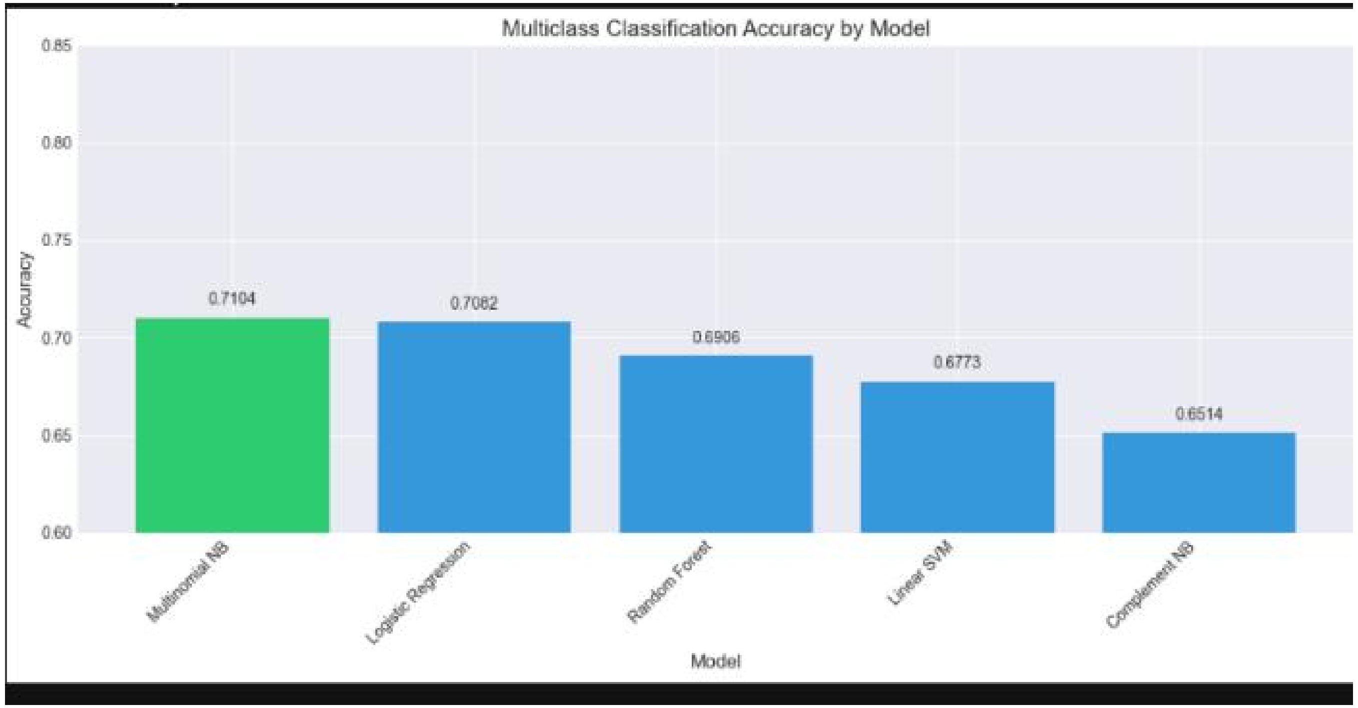
Binary confusion Matrix



interpretation

- High Positive recall - correctly predicting 587/594 actual positives
- High false positive rate - it struggles significantly with the negative samples.
- model is biased toward predicting the positive class

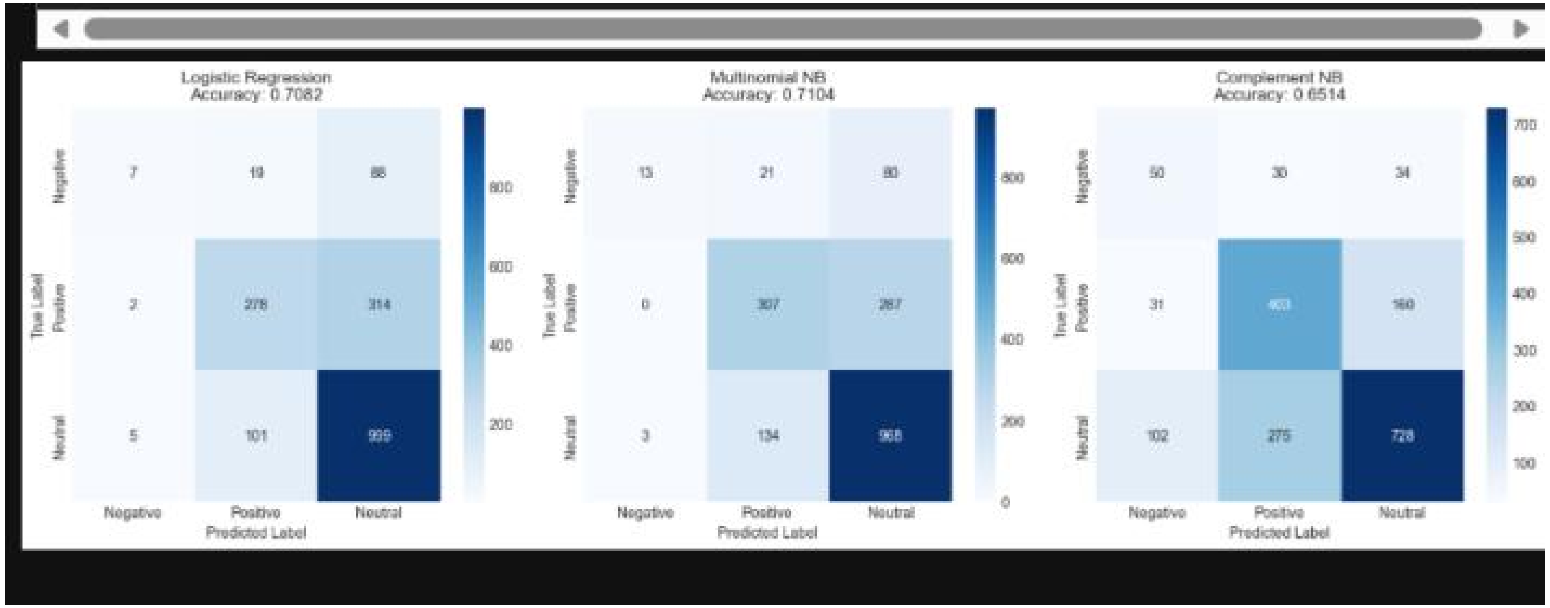
Multiclass Results



Interpretation

- MultinomialNB achieved the highest overall accuracy
- performance drop - this was experienced across all models (this means that introducing the neutrall class significantly increased the difficulty in classification)

Multiclass Confusion Matrix

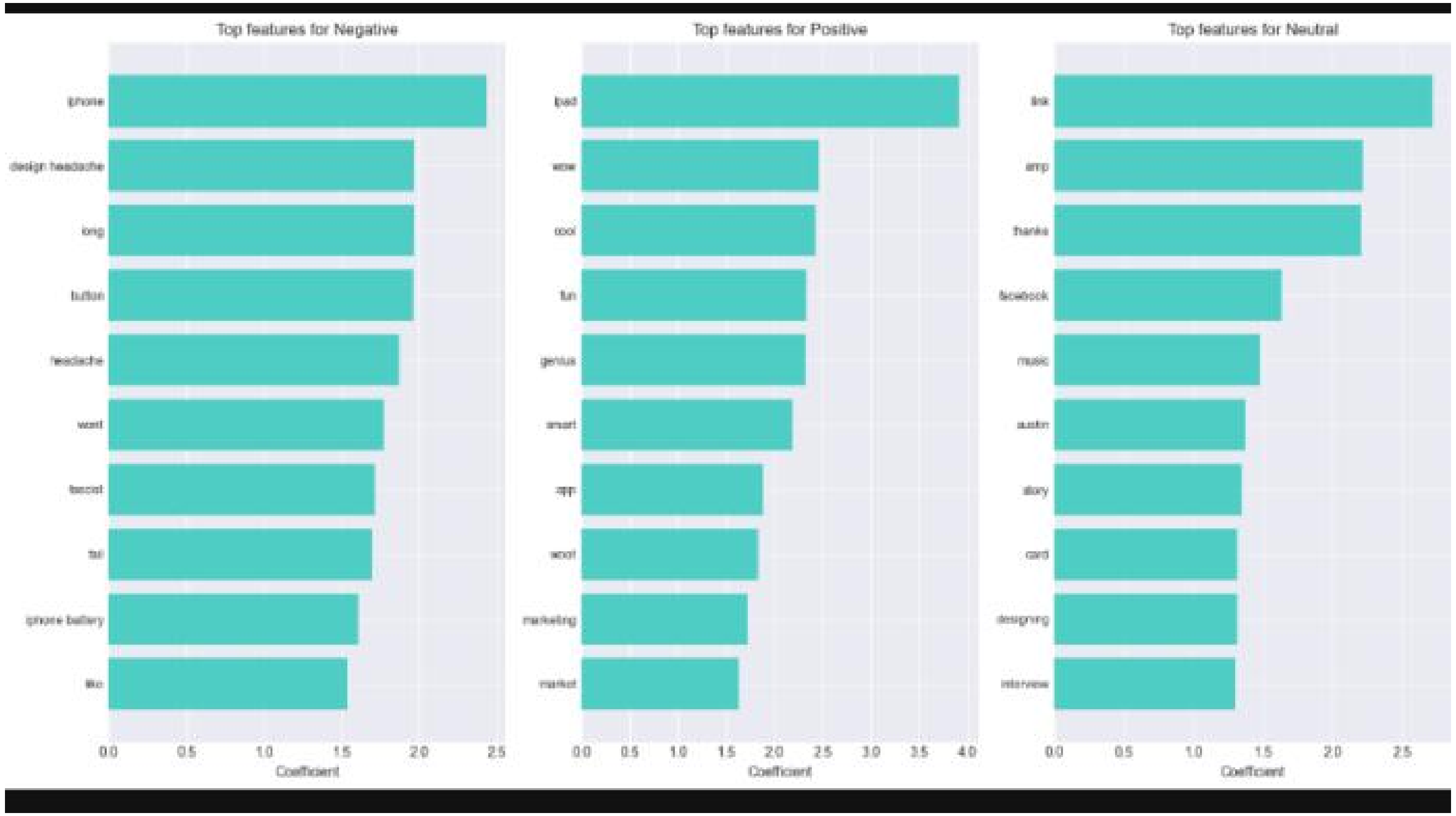


Insights from the top 3 models

- neutral often misclassified as positive/negative
- positive tweets are easier to classify
- MultinomialNB is still the best of all the multiclass models

NB - this is expected for sentiment tasks

Feature Importance



Business Recommendations

IMMEDIATE (Week 1):

- • Deploy Random Forest model for positive vs negative and MultinomialNB for positive/negative/neutral
- • Set real-time alerts for negative mentions
- • Monitor top products: iPad, iPhone, Apple

SHORT-TERM (Month 1):

- • Add confidence threshold (flag <0.7)
- • Retrain model weekly
- • Create product-specific models

EXPECTED ROI:

- • 90% reduction in manual monitoring
- • 24/7 brand sentiment coverage
- • Minute-level crisis detection

Key Takeaways

- 1. Traditional ML works great
- random forest achieved 87.85% accuracy, providing excellent recall for positive sentiment.
- No deep learning needed for this data

- 2. Neutral tweets are the challenge
- Hardest to classify correctly
- Often mistaken for positive/negative
- accuracy drops

- 3. Apple dominated SXSW conversation
- 40%+ of tweets mentioned iPad/iPhone
- Pop-up store generated massive buzz

- 4. Model is production-ready
- deploy MultinomialNB for balanced low latency text classification
- Fast inference (<1ms per tweet)
- Interpretable and balanced

Next Steps

IMMEDIATE:

- 1. Deploy model as API
- 2. Create monitoring dashboard
- 3. Set up negative alerting

FUTURE IMPROVEMENTS:

- • Collect more neutral tweet data
- • Add VADER sentiment scores
- • Experiment with XGBoost
- • Build time-series trends

- THANK YOU!